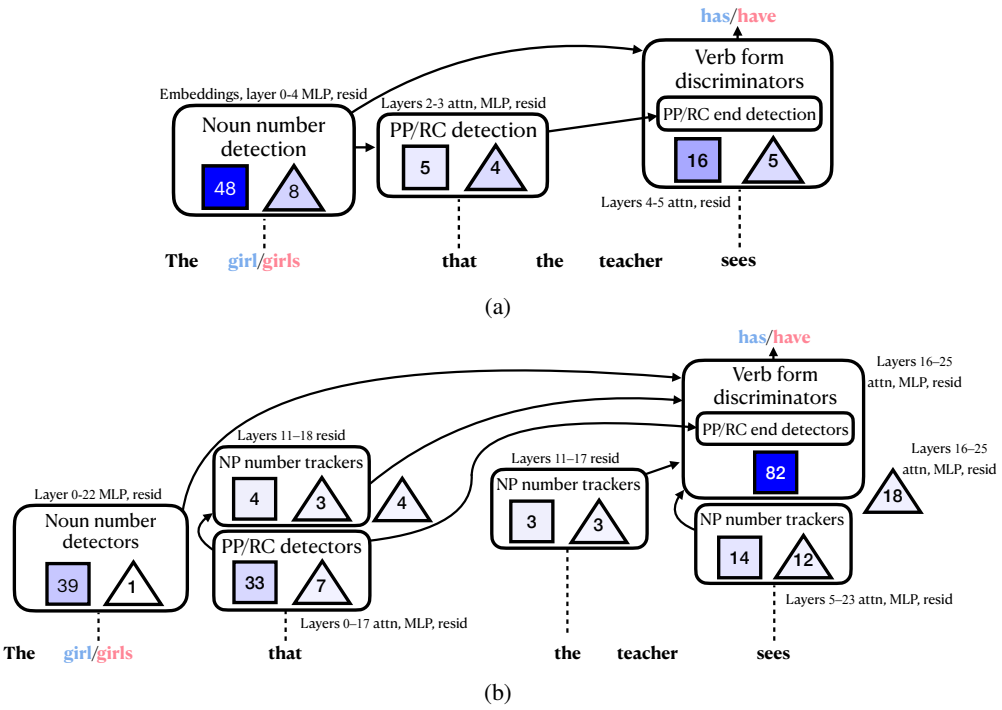

Figure 4: Summary of Pythia's (a) and Gemma 2's (b) circuits for agreement across RC (full circuits in App. C.1). The models detect the number of the subject. Then, they detect the start of a PP/RC modifying the subject. Verb form discriminators promote particular verb inflections (singular or plural). Gemma 2 additionally uses separate features to track the number of the noun that heads the current noun phrase. Squares show number of feature nodes in the group and triangles show number of SAE error nodes, with the shading indicating the sum of $\hat{\text{IE}}$ terms across nodes in the group. As we cannot directly interpret the triangles, we rely on their positions or inclusion in other groups to label them. If the label is ambiguous, we leave the triangles outside the boxes.

which disallow this assumption. For example, inputs for different classes might come from different data sources (Zech et al., 2018). Additionally, some have raised concerns (Ngo et al., 2024; Casper et al., 2023) that sophisticated LMs trained with human feedback (Christiano et al., 2023) in settings with easy-to-hard domain shift (Burns et al., 2023; Hase et al., 2024) will be misaligned because, in these settings, "overseer approval" and "desirable behavior" are equally predictive of training reward labels. More fundamentally, the problem with unintended signals is that they are *unintended*—not they are insufficiently predictive—and we would like our methods to reflect this.

We thus propose Spurious Human-interpretable Feature Trimming (SHIFT), where a human inspects a classifier's feature circuit and removes features which they judge to be task-irrelevant. We show that SHIFT removes sensitivity to unintended signals without access to disambiguating labeled data, or even without knowing what the signals are ahead of time.

Method. Suppose we are given labeled training data $\mathcal{D} = \{(x_i, y_i)\}$; an LM-based classifier C trained on $\mathcal{D}$; and SAEs for various components of C. To perform SHIFT, we:

1. Apply the methods from §3 to compute a feature circuit that explains C's accuracy on inputs $(x, y) \sim \mathcal{D}$ (e.g., using metric $m = -\log C(y|x)$).
2. Manually inspect and evaluate for task-relevancy each feature in the circuit from Step 1.
3. Ablate from C features judged to be task-irrelevant to obtain a classifier C'.
4. (Optional) Further fine-tune C' on data from $\mathcal{D}$.

Step 3 removes the classifier's dependence on unintended signals we can identify, but may disrupt the classifier's performance for the intended signal. Step 4 can be used to restore some performance.

Method	Pythia-70M			Gemma-2-2B		
	↑Profession	↓Gender	↑Worst group	↑Profession	↓Gender	↑Worst group
Original	61.9	87.4	24.4	67.7	81.9	18.2
CBP	83.3	60.1	67.7	90.2	**50.1**	86.7
Random	61.8	87.5	24.4	67.3	82.3	18.0
SHIFT	88.5	54.0	76.0	76.0	51.5	50.0
SHIFT + retrain	**93.1**	**52.0**	**89.0**	**95.0**	52.4	**92.9**
Neuron skyline	75.5	73.2	41.5	65.1	84.3	5.6
Feature skyline	88.5	54.3	62.9	80.8	53.7	56.7
Oracle	93.0	49.4	91.9	95.0	50.6	93.1

Table 2: Accuracies on balanced data for the intended label (profession) and unintended label (gender). "Worst group accuracy" refers to whichever profession accuracy is lowest among male professors, male nurses, female professors, female nurses.

Experimental setup. We illustrate SHIFT using the Bias in Bios dataset (BiB; De-Arteaga et al., 2019). BiB consists of professional biographies, and the task is to classify an individual's profession based on their biography. BiB also provides labels for a spurious feature: gender. We subsample BiB to produce two sets of labeled data:

- The **ambiguous set**, consisting of bios of male professors (labeled 0) and female nurses (labeled 1).
- The **balanced set**, consisting of an equal number of bios for male professors, male nurses, female professors, and female nurses. These data carry **profession labels** (the intended signal) and **gender labels** (the unintended signal).

The **ambiguous set** represents a worst-case scenario: the unintended signal is perfectly predictive of training labels. Given only access to the **ambiguous set**, our task is to produce a **profession** classifier which is accurate on the **balanced set**.

We adapt Pythia-70M and Gemma-2-2B into classifiers by training linear classification heads with the **ambiguous set**; see App. E.1 for probe training details. We then discover feature circuits for these classifiers using the zero-ablation variant described in §3.1; the Pythia circuit contains 67 features, and the Gemma circuit contains 46. We manually interpret each feature using the Neuronpedia interface (Lin & Bloom, 2023), which displays maximally activating dataset exemplars on a large text corpus, as the features' direct effects on output logits. We judge 55 of the Pythia features and 43 of the Gemma features to be task-irrelevant—e.g., features that promote female-associated language in biographies of women, as in Figure 19 (see App. D for more examples features). Although this interpretability step uses additional unlabeled data, we emphasize that we never use additional *labeled* data (or even additional unlabeled classification data).

To apply SHIFT, we zero-ablate these irrelevant features. Finally, we retrain the linear classification head with the **ambiguous set** using activations extracted from the ablated model. We evaluate all accuracies on the **balanced set**.

Baselines and skylines. To contextualize the performance of SHIFT, we also implement:

- **SHIFT with neurons.** Perform SHIFT, but using neurons instead of SAE features.
- **Concept Bottleneck Probing** (CBP), adapted from Yan et al. (2023) (originally for multimodal text/image models). CBP works by training a probe to classify inputs x given access only to a vector of affinities between the LM's representation of x and various concept vectors. See App. E.2 for implementation details.
- **Random feature ablations.** Perform SHIFT, but using (the same number of) randomly selected SAE features instead of features selected by a human annotator.
- **Feature skyline.** Instead of relying on human judgement to evaluate whether a feature should be ablated, we zero-ablate the 55 (for Pythia) or 43 (for Gemma) features from our circuit that are most causally implicated in **spurious feature** accuracy on the **balanced set**.
- **Neuron skyline.** The same as the feature skyline, but mean-ablating 55 or 43 neurons.
- **Oracle.** A classifier trained on **ground-truth labels** on the **balanced set**.

Results. We find (Table 2) that SHIFT almost completely removes the classifiers' dependence on gender information for both models. In the case of Gemma (but not Pythia), the feature ablations

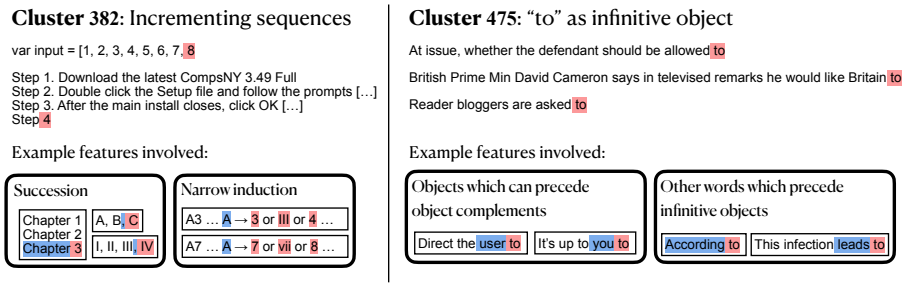

Figure 5: Example clusters and features which participate in their circuits (see App. C.3 for the full circuits). Features are active on tokens shaded in blue and promote tokens shaded in red. *(left)* An example *narrow* induction feature recognizes the pattern $A3 \ldots A$ and copies information from the 3 token. This composes with a succession feature to implement the prediction $A3 \ldots A \to 4$. *(right)* One feature promotes "to" after words which can take infinitive objects. A separate feature activates on objects of verbs or prepositions and promotes "to" as an object complement.

damage model performance; however, this performance is restored (without reintroducing the bias) by further training on the ambiguous set. Comparing SHIFT without retraining to the feature skyline, we further observe that SHIFT optimally or near-optimally identifies the best features to remove.

SHIFT critically relies on the use of properly selected SAE features. When ablating random SAE features, we see essentially no effect on probe performance. When applying SHIFT with neurons, essentially none of the neurons are interpretable, making it difficult to tell if they ought to be ablated; see Appendix D for examples. Because of this, we abandon the SHIFT with neurons baseline. Even using the **balanced set** to automatically select neurons for removal (the neuron skyline) fails to match SHIFT's performance, as the neurons most implicated in **spurious feature** classification are also useful for **ground-truth** classification.

5 UNSUPERVISED CIRCUIT DISCOVERY AT SCALE

Previous work on circuit analysis relied on human-collected datasets to specify LM behaviors (Wang et al., 2023; Conmy et al., 2023; Hanna et al., 2023). However, LMs implement numerous interesting behaviors, many of which may be counterintuitive to humans. In this section, we adapt our techniques to produce a near-fully-automated interpretability pipeline, starting from a large text corpus—here, a large subset of The Pile (Gao et al., 2020)—and ending with thousands of feature circuits for auto-discovered model behaviors. These experiments are performed with Pythia-70M.

We proceed in two steps:

1. **Behavior discovery via clustering.** We interpret our large text corpus as a dataset $\{(x_i, y_i)\}$ of contexts x_i with ground-truth next tokens y_i. Following Michaud et al. (2023), we associate a vector $\mathbf{v}_i = \mathbf{v}(x_i, y_i)$ to each sample and apply a clustering algorithm to $\{\mathbf{v}_i\}$; this segments our large corpus into a number of smaller subcorpora corresponding to the clusters. Although this approach is entirely unsupervised, many of the resulting subcorpora capture human-interpretable model behaviors, such as predicting the next number in a sequence (Fig 5). We experiment with a number of ways of assigning $(x_i, y_i) \mapsto \mathbf{v}_i$, such as using the training gradient $\nabla_\theta \log P_\theta(y_i|x_i)$ as in Michaud et al. (2023) as well as approaches which leverage SAE activations or gradients. See App. G for details.
2. **Circuit discovery.** Given a subcorpus $\mathcal{D} = \{(x_i, y_i)\}$, we apply the zero-ablation variant of our feature circuit discovery technique from §3 using the dataset $\mathcal{D}$ and metric $m = -\log P(y_i|x_i)$. Thus, to each subcorpus we associate a feature circuit.

We present example clusters, as well as interesting features participating in their associated circuits (Figure 5). An interface for exploring all of our clusters and (unlabeled) circuits can be found at `feature-circuits.xyz`.

While evaluating these clusters and circuits is an important open problem, we generally find that these clusters expose interesting LM behaviors, and that their respective feature circuits can provide